SCENTINEL

OBP candidate selection · analysis report

iGEM URV 2026

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Target VOC       :  methyl salicylate
CAS              :  119-36-8
Top N candidates :  8
Interferents     :  - none -
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tau (selectivity):  5
w1 affinity      :  0.450
w2 selectivity   :  0.250
w4 stability     :  0.150
w5 promiscuity   :  0.150
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SMAA configuration

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Method 2 (Perturbation):  c = 50%,  500 iterations
Method 3 (pure SMAA)   :  500 iterations
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SCENTINEL · Detailed results | VOC: methyl salicylate

#	OBP	Conserv.	Conf.	c2	s1	s2	s4	s5	Ki (uM)	Interf	Danger	Type
1	DcitOBP3	0.4919	0.74	0.00	0.472	0.500	1.000	0.902	7.01	-	-	Classic OBP
2	AcerOBP11	0.4833	0.74	0.00	0.479	0.500	1.000	0.814	6.83	-	-	Classic OBP
3	AcerOBP2	0.4657	0.73	0.00	0.427	0.500	1.000	0.888	8.29	-	-	Classic OBP
4	GmolOBP7	0.4632	0.75	0.00	0.381	0.500	1.000	0.964	9.81	-	-	Classic OBP
5	GmolPBP2	0.4452	0.74	0.00	0.452	0.500	0.850	0.793	7.55	-	-	PBP
6	PxylGOBP2	0.4399	0.74	0.00	0.437	0.500	0.850	0.795	7.97	-	-	GOBP2
7	SmosOBP12	0.4361	0.74	0.00	0.386	0.500	1.000	0.780	9.62	-	-	Classic OBP
8	RpadOBP3	0.4351	0.72	0.00	0.350	0.500	1.000	0.951	11.00	-	-	Classic OBP

Conserv. = Score x Confidence · c2 = selectivity confidence · Interf = measured/total · Danger = vineyard interferents binding as strong as the target
Row tint: green = confidence >= 0.80 lilac = 0.50-0.80 pink = < 0.50

Dangerous vineyard interferents per candidate

(Ki_interf <= Ki_target x 1.33; v = binds stronger than the target)

OK No candidate in the top has measured vineyard interferents binding as strong as the target.

SCENTINEL · s5 superbinders | VOC: methyl salicylate

Superbinders - VOCs (any, not only vineyard) bound strongly

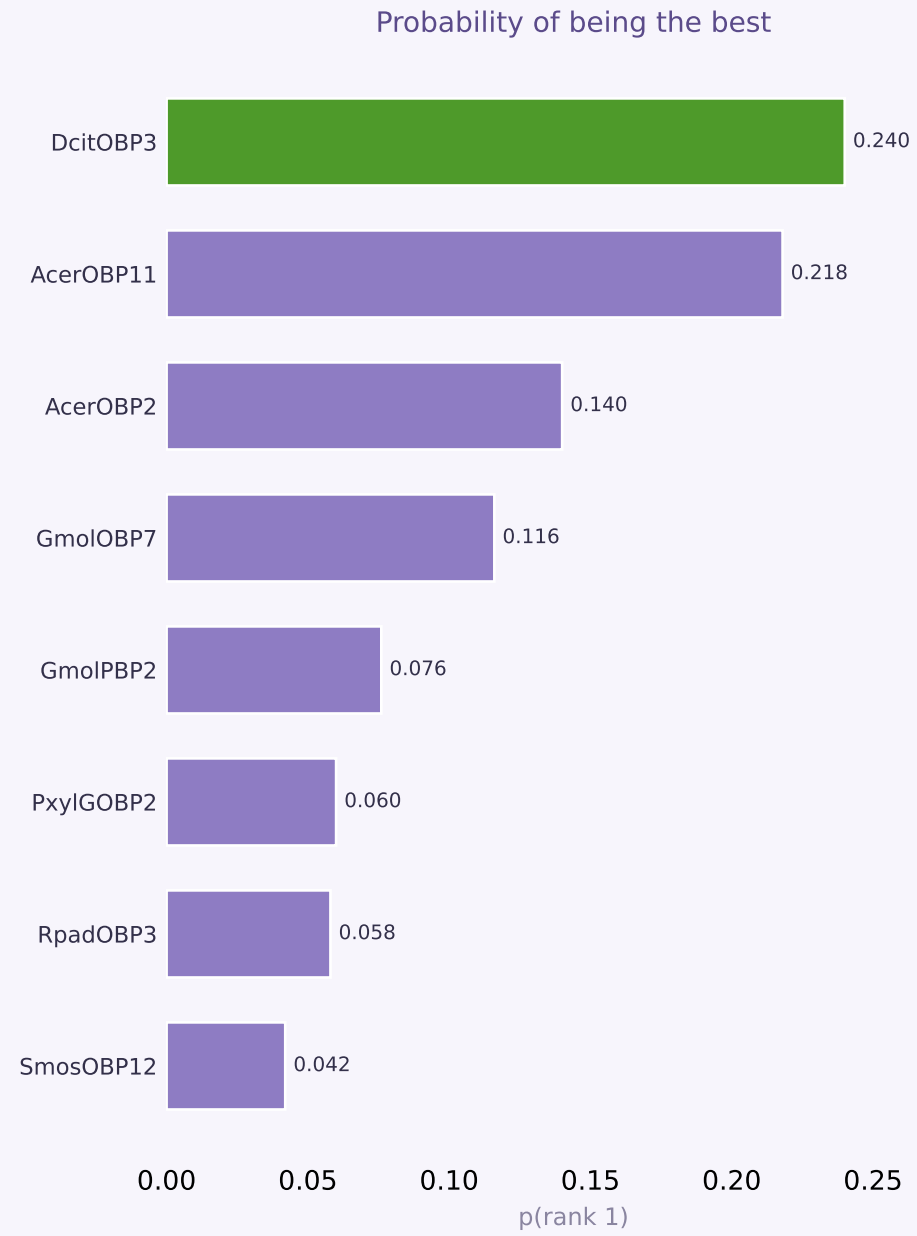
* = Ki <= 2.0 uM (strong) v = binds stronger than the target [n] = total superbinders

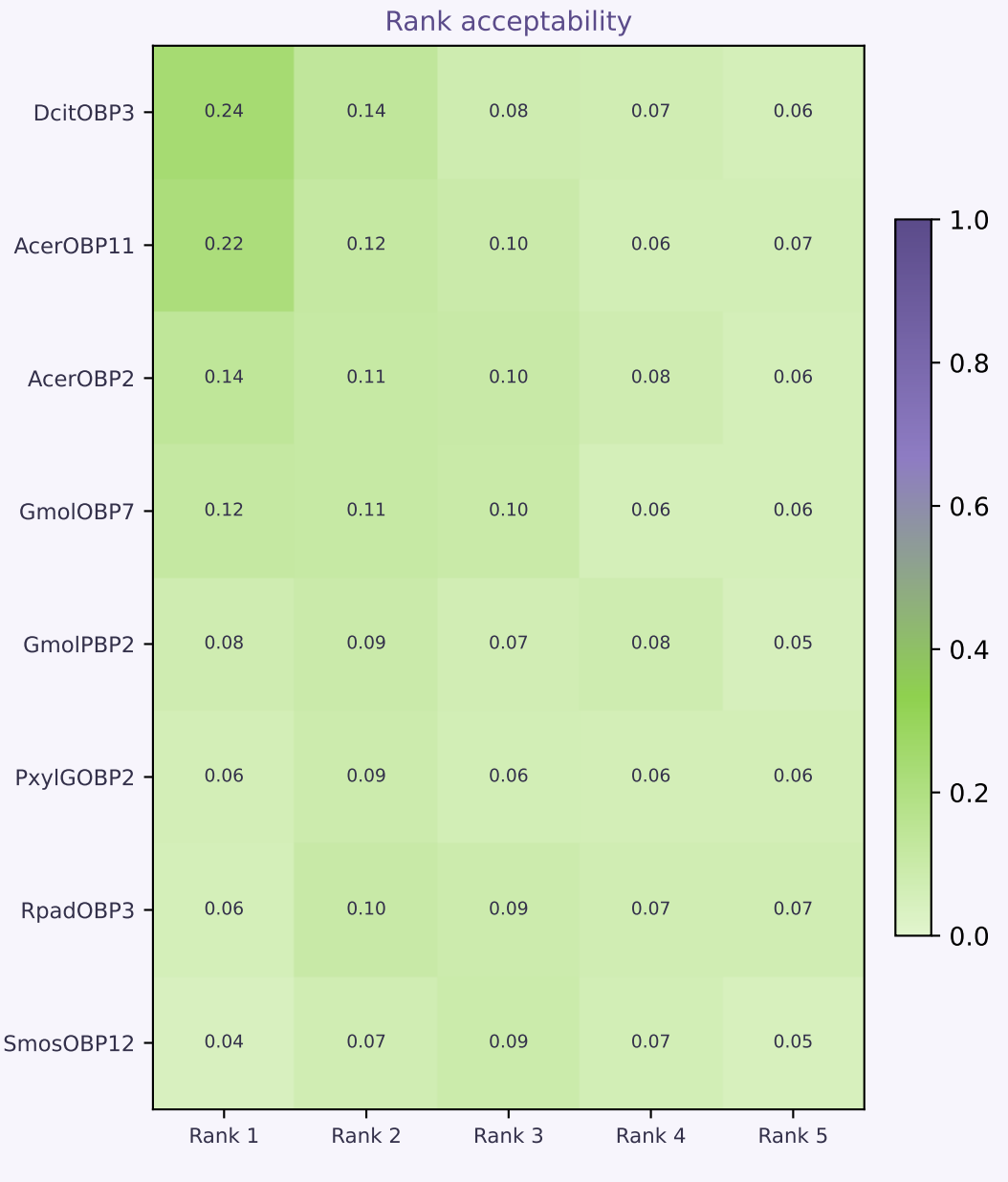
1. DcitOBP3 [11] DL linalool(3.5v), dimethyl trisulfide(4.2v), C9:Ald(4.3v), p-cymene(4.7v), (1R)-(+)-alpha-pinene(5.0v), (1S)-(-)-alpha-pinene(5.1v), dimethyl disulfide(5.1v), acetic acid(5.3v), trans-(3E)-beta-ocimene(5.9v), lauric acid(6.4v) +1
2. AcerOBP11 [16] HOB(1.4*v), Methyleugenol(2.7v), 1-hexanol(2.8v), isoamyl acetate(3.3v), phenylethanal(3.8v), ionone (beta)(3.8v), Ethyl hexadecanoate(3.8v), eucalyptol(4.3v), 2-heptanone(4.3v), l-(-)-Menthol(4.3v) +6
3. AcerOBP2 [4] phenylethanal(2.7v), ionone (beta)(4.1v), geraniol(5.5v), HOB(7.6v)
4. GmolOBP7 [4] Pear ester(2.5v), 12:Ald(4.3v), 12:OH(7.5v), alpha-ocimene(7.8v)
5. GmolPBP2 [8] Z8-12:Ac(1.1*v), E8-12:AC(1.1*v), Z8-12:OH(2.0*v), alpha-pinene(2.0v), 12:OH(2.8v), heptanal(3.8v), Cis-3-Hexenyl acetate(5.8v), ionone (beta)(6.7v)
6. PxylGOBP2 [17] Farnesene(1.0*v), (Z)-3-Hexenyl isovalerate(2.4v), Z11-16:Ac(2.4v), caryophyllene(2.4v), Z11-16:OH(2.7v), Z11-16:Ald(2.9v), phenylethanal(4.2v), DL linalool(4.5v), benzoic acid(4.9v), geraniol(5.5v) +7
7. SmosOBP12 [16] 3-hexanol(3.3v), heptyl acetate(3.3v), 3-Carene(3.8v), D-panthenol(4.1v), Pentyl pentanoate(4.2v), 1-undecanol(4.4v), 1-octen-3-ol(5.3v), propanoic acid, hexyl ester(5.9v), alpha-pinene(6.2v), 3-hexanone(6.3v) +6
8. RpadOBP3 [4] beta-farnesene(8.9v), Z-3-hexen-1-ol(9.2v), (1S)-(-)-alpha-pinene(9.7v), r-limonene(10.5v)

SCENTINEL · Experimental vs Computational | VOC: methyl salicylate

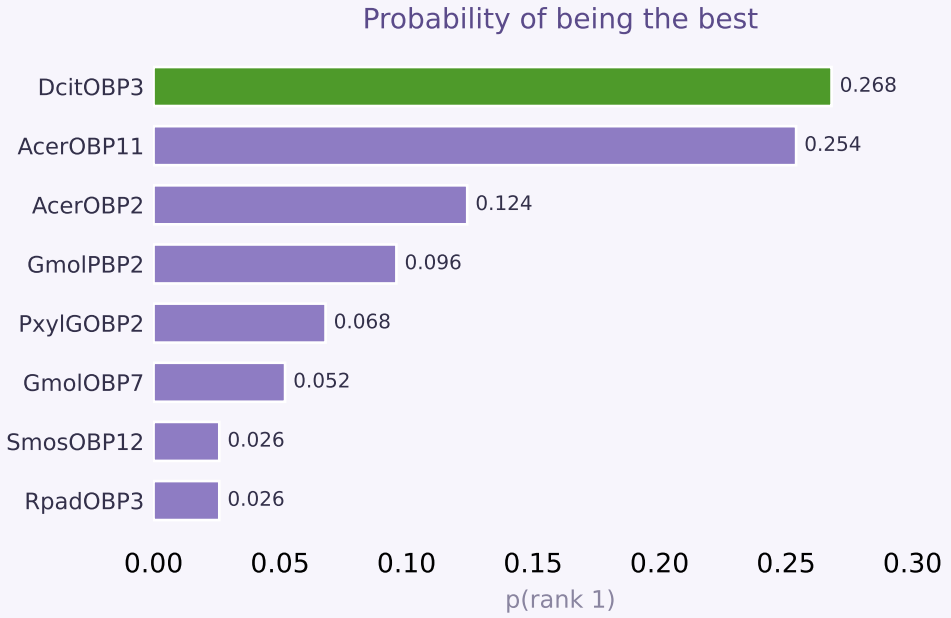


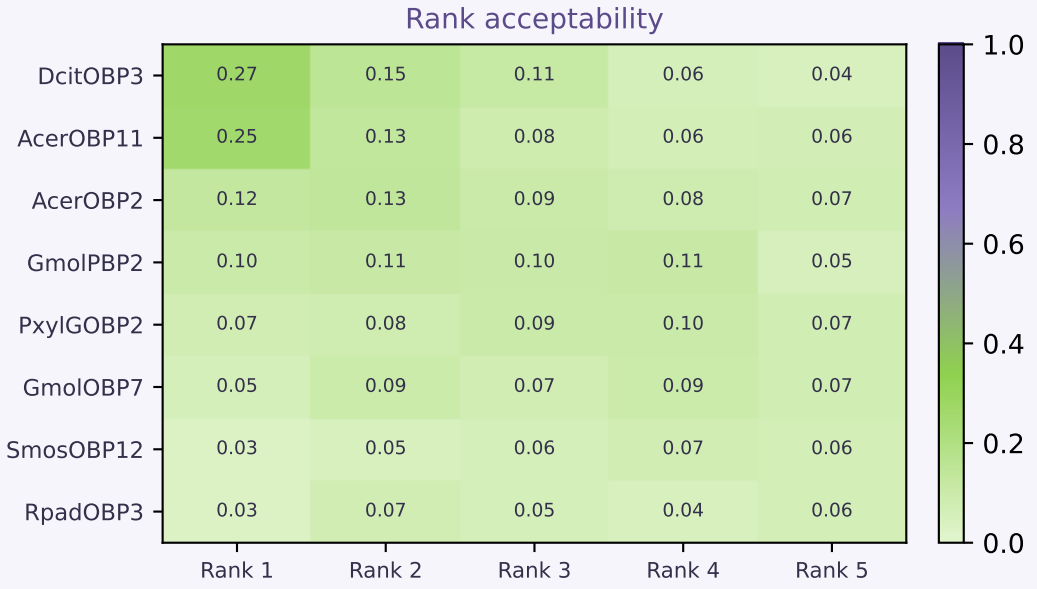
SCENTINEL · Method 2 - Weight Perturbation | VOC: methyl salicylate

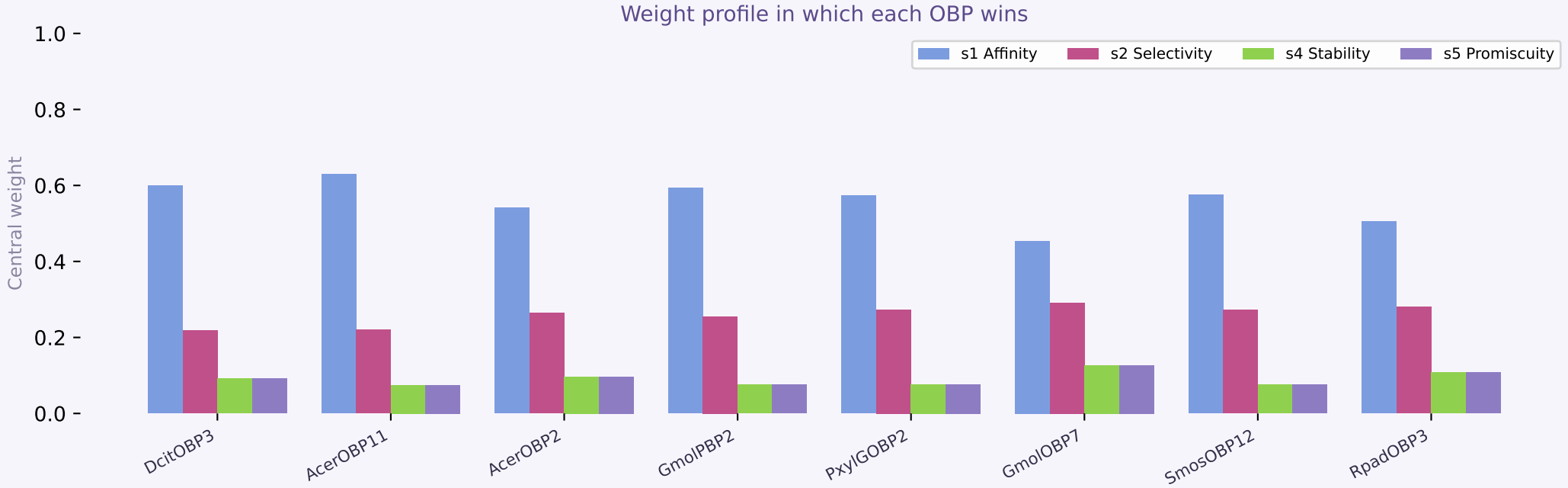


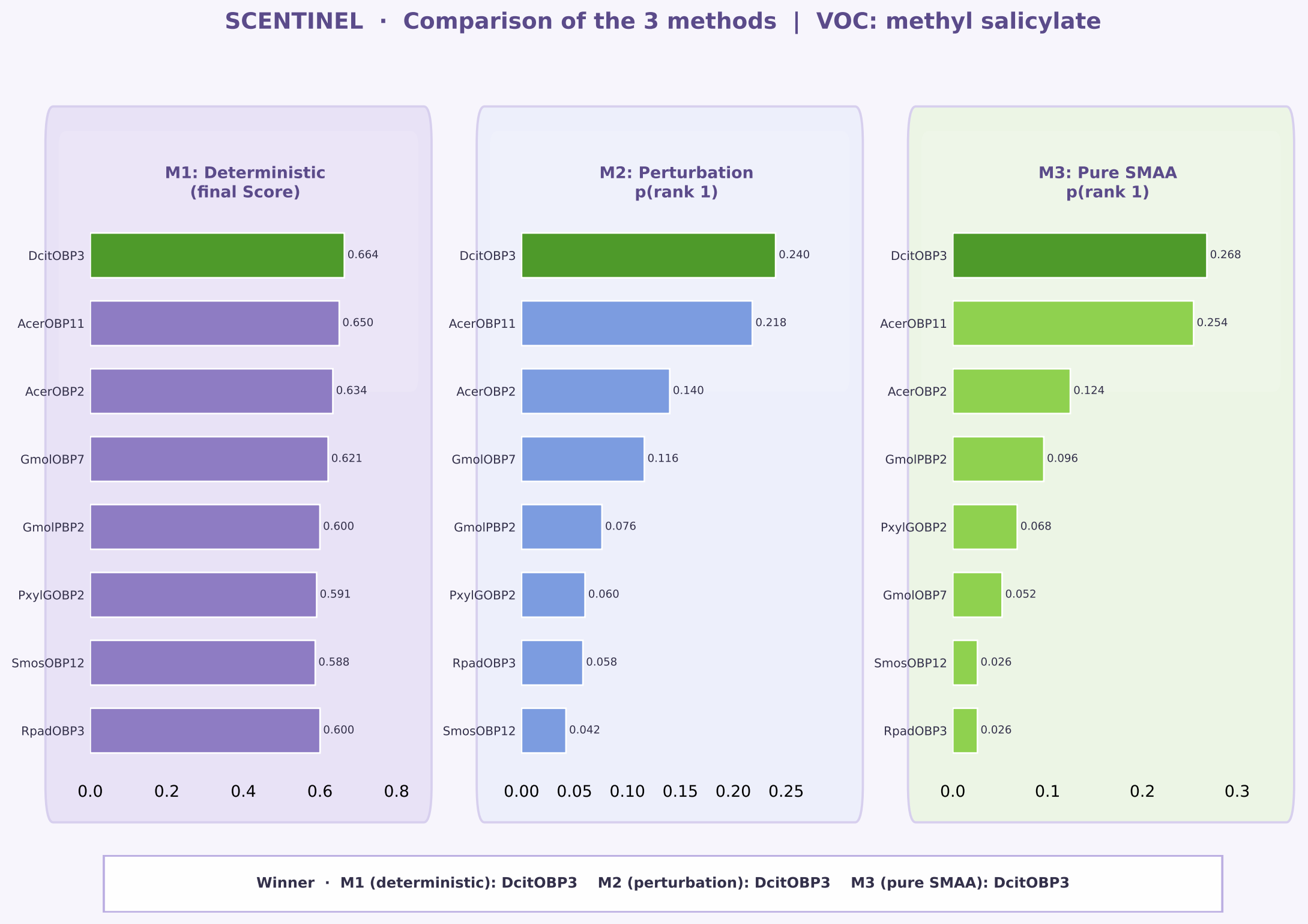


SCENTINEL · Method 3 - Pure SMAA | VOC: methyl salicylate








SCENTINEL · Comparison of the 3 methods | VOC: methyl salicylate
M1: Deterministic (final Score)
DcitOBP3 0.664
AcerOBP11 0.650
AcerOBP2 0.634
GmolOBP7 0.621
GmolPBP2 0.600
PxylGOBP2 0.591
SmosOBP12 0.588
RpadOBP3 0.600
0.0 0.2 0.4 0.6 0.8
M2: Perturbation p(rank 1)
DcitOBP3 0.240
AcerOBP11 0.218
AcerOBP2 0.140
GmolOBP7 0.116
GmolPBP2 0.076
PxylGOBP2 0.060
RpadOBP3 0.058
SmosOBP12 0.042
0.00 0.05 0.10 0.15 0.20 0.25
M3: Pure SMAA p(rank 1)
DcitOBP3 0.268
AcerOBP11 0.254
AcerOBP2 0.124
GmolPBP2 0.096
PxylGOBP2 0.068
GmolOBP7 0.052
SmosOBP12 0.026
RpadOBP3 0.026
0.0 0.1 0.2 0.3
Winner · M1 (deterministic): DcitOBP3 M2 (perturbation): DcitOBP3 M3 (pure SMAA): DcitOBP3